HACKATHON 6 · LIVE PULSE · SDG 9

Maintainer Pressure Meter

A live sentiment meter on the GitHub comments of the open-source projects our digital infrastructure runs on.

Yassir Balah · yassirdh.github.io/maintainer-druk-meter

THE PROBLEM

A tired maintainer is an attack surface

> “Progress will not happen until there is new maintainer.”
>
> Unknown account to the sole xz maintainer, June 2022

> “My ability to care has been fairly limited mostly due to longterm mental health issues.”
>
> Lasse Collin, xz maintainer, replying

> “Why can’t you commit this yourself?”
>
> Same account, a week later

2024

A backdoor (CVE-2024-3094) in xz, planted by the “helper” who took over, caught just before stable Linux releases.

58%

of maintainers have quit or considered quitting a project.

Sources: Russ Cox, xz timeline (research.swtch.com) · Tidelift 2023 State of the Open Source Maintainer

USER AND DECISION

Built for funds that support maintainers

Who reads it

A programme officer at the Sovereign Tech Agency (Germany) or NLnet (the Netherlands).

What they decide

Each week: which project to contact or offer support: a fellowship, an extra maintainer, help with moderation.

What the meter does

Points to where a human should start reading. It decides nothing on its own.

Who is missing from the stream

Mailing lists, chat and private email · non-English speakers · people who already gave up · xz and zlib: 0 GitHub comments in 48 hours

HOW IT WORKS

Every 15 minutes, on its own

1

Fetch

GitHub REST API, new comments on 15 critical projects

2

Clean

Drop bots, code and logs; @names become @user; authors become a salted hash

3

Score

twitter-roberta sentiment model from Hugging Face, run in my own code

4

Store

comments.csv with timestamps, committed by GitHub Actions

5

Show

Dashboard on GitHub Pages, always up to date

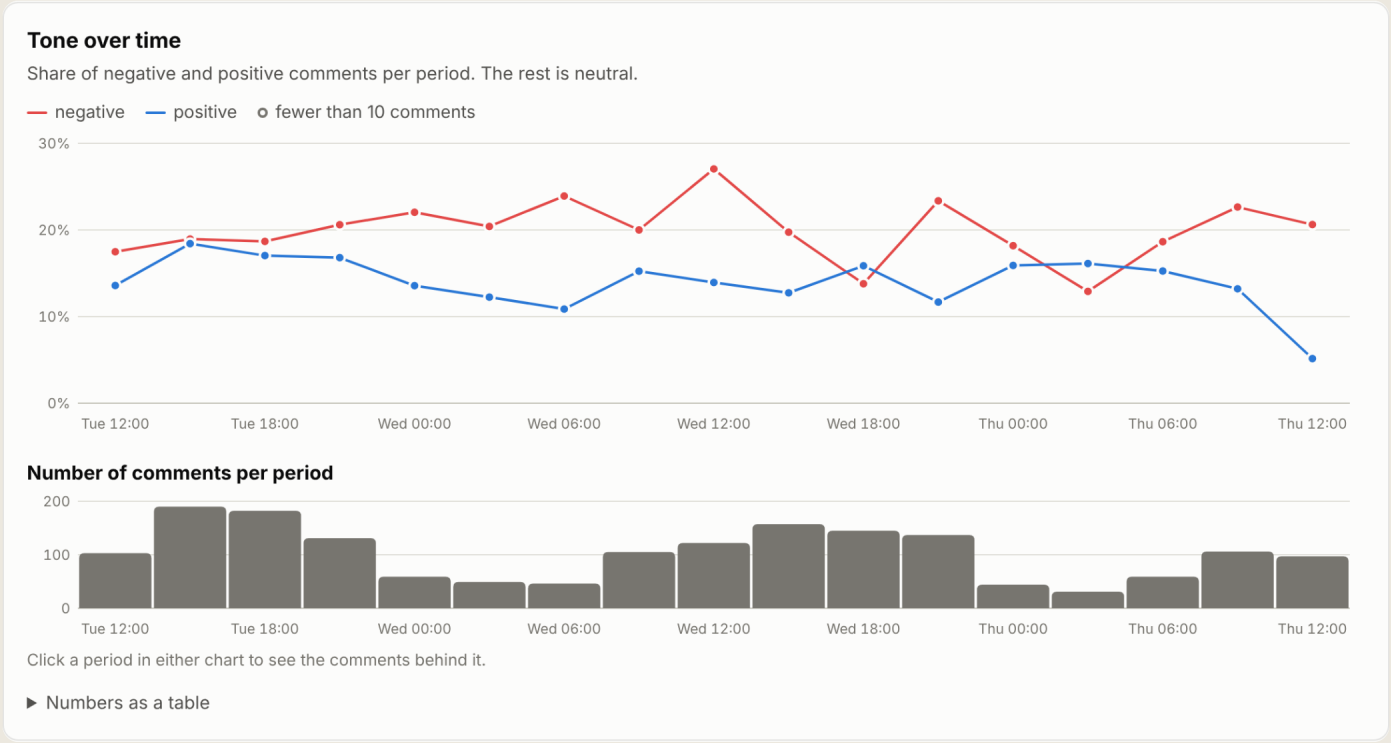
50+ h
running continuously

1,760+
comments analysed

15
critical projects

LIVE DEMO

What’s behind Wednesday’s peak?



27%

negative, Wed 12:00–15:00 (122 comments)

Home Assistant users: things stopped working after an update

Maintainers closing AI-generated PRs: “fully vibecode… against our AI Policy”

Live: yassirdh.github.io/maintainer-druk-meter

HONEST TEST · 50 COMMENTS, LABELLED BLIND

Model 1 agrees with me 72% of the time

Method	Agrees with me	Negatives caught	Negative alarms correct
Model 1 · twitter-roberta (live)	72%	10 of 14	77%
Model 2 · xlm-roberta (multilingual)	64%	11 of 14	55%
Word list (no AI)	38%	4 of 14	36%
Always say “neutral”	54%	0 of 14	–

My labels: 14 negative · 27 neutral · 9 positive. Weak spot: model 1 found only 3 of my 9 positive comments. Small test: one comment is 7 points of recall.

WHERE IT GOES WRONG

It reads the words, not the tone

Jargon becomes anger

“Behavior 2 seems likely to cause issues for code like this…”

Me: neutral
Model 1: **negative** (74%)

Thanks plus a bug

“Thanks for spotting this! … it’s a regression since 10.2.3.”

Me: positive
Model 1: **neutral** (64%)

Polite frustration

“…getting a real pita 😅 … a simple filter would do wonders.”

Me: negative
Model 1: **neutral** · Model 2: **positive**

Three of the four negatives model 1 missed were polite requests. Developers rarely shout.

THE FAILURE THAT TAUGHT ME THE MOST

The model was ready.
The data source was not.

5 of 6

real 2022 pressure messages to the xz maintainer that model 1 flags as negative.

0 of 6

that my meter would have seen: they were sent on a mailing list, not on GitHub.

Italian

The one Italian comment: right by the English model, wrong by the multilingual one.

DECISION AND ETHICS

Model 1, as a pointer, not a judge

Why model 1

- 77% of its negative alarms are right (model 2: 55%)
- Model 2: 9 false alarms instead of 3, for 1 extra real negative
- A busy fund officer stops trusting a noisy dashboard
- The word list is worse than always saying “neutral”

What I did about the risks

- No usernames: @user, authors as a salted hash
- A count beside every %; under 10 comments: hollow dot
- A flag when one account drives the line
- The comments are one click behind every peak

Not for: judging people, automatic funding decisions, or “toxic project” rankings. Quiet is not the same as healthy.